DE LA SALLE UNIVERSITY
College of Computer Studies
Department of Computer Technology

Integrative Programming
Machine Project Proposal & Specification

Introduction

This machine project serves as a culminating requirement for the Integrative Programming course. It is designed to assess students' ability to integrate multiple web development technologies—such as HTML for structure, CSS for presentation, PHP for server-side logic, and databases for data management—into a cohesive and functional web-based system. The project emphasizes proper coding practices, separation of concerns, and the practical application of programming concepts discussed throughout the course. Students are expected to demonstrate not only technical proficiency but also the ability to design systems that address real-world problems in an organized and systematic manner.

The system to be developed must be aligned with the operational needs of a specific organization or institution, whether real or hypothetical. Each project group is required to identify an organizational context (e.g., school, small business, clinic, retail store, government office, or service-oriented organization) and **clearly define the problem or inefficiency that the proposed web-based system aims to address.**

Although having an actual client is not required, the proposed solution must be realistic, relevant, and grounded in real-world scenarios. The design and functionality of the system should demonstrate how it can effectively support organizational processes, improve efficiency, enhance data management, or provide better services to its intended users.

The client-side system should focus on meeting the needs of end users within the organization, while the server-side (admin) system should support administrative tasks such as data management, monitoring, reporting, and system configuration. Together, both systems must reflect a coherent solution to the identified organizational problem.

Project Proposal Submission and Presentation

Students must use the provided template to clearly describe and define the specifications of their proposed project.

- This group project is composed of four (4) members only.
- The tentative Project Proposal Presentation will be held on Week 4.
- During the proposal presentation, the project scope and features will be discussed with the instructor.
- After the presentation, groups are required to revise and resubmit their

proposal.
- The resubmitted proposal (due one week after the presentation) will be considered the final and approved project specification.

Development Tools and Technologies

- HTML & CSS – for structure and styling
- PHP – for server-side logic
- MySQL – for database management
- JSON and XML may be used for data exchange and integration
- JavaScript (Front-end only)
    - UI behavior and interactivity
    - DOM manipulation
    - Form validation
    - AJAX-based data fetching

Important: The use of JavaScript for server-side development (e.g., Node.js, Express, backend APIs, or business logic) is strictly prohibited.

Integration Tasks

- The client-side and server-side systems must be properly integrated using any appropriate method (e.g., data integration or application-level integration).
- Data between the two systems must be consistent, synchronized, and updated correctly. Any changes made through the admin system should be reflected accurately on the client-side system.

Final Project Submission Guidelines

Each group must submit the following:

- Complete Source Code
- Entity Relationship Diagram (ERD)
- Exported Database File (sql)

Grading and Presentation Guidelines

- The Final Machine Project Presentation will be held on Week 14.
- Each group will be given 30 minutes, inclusive of:
    - System demonstration
    - Explanation of features
    - Question and Answer portion
- The project will be evaluated based on the approved proposal. Any feature not included in the final proposal may not be considered during grading.
- Grading will follow the official rubric provided by the instructor.

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Group Name: _____ Section: _____

Group Members: 1. _____
2. _____
3. _____
4. _____

Project Overview

1.1 Project Title

Provide a clear and descriptive project title.

1.2 Project Description

Briefly describe what the system does, its purpose, and the problem it solves (2–3 paragraphs). Focus on the idea, not the technical implementation

1.3 Target Users

Who will use the system?

- *End Users (e.g., customers, students, employees)*
- *Administrator(s)*
- *others*

Proposed System Features

List the features you PLAN to implement.
Separate Client Side and Server/Admin Side features.

Part 1: Client Side

Features accessible to normal users / customers. Should discuss its capabilities, and limitations.

Part 2: Server Side

Features accessible only to administrators. Should discuss its capabilities, and limitations.

Criteria	Ratings	Pts
Overall System Integration	• **20 pts - Full Marks:** Achieves seamless integration across all components with exceptional performance and user experience • **16 pts - Proficient:** Good overall integration with minor issues in performance or user experience • **12 pts - Competent:** Adequate integration with noticeable limitation in some components • **8 pts - Developing:** Poor integration with significant issues affecting performance or user experience • **4 pts - Beginning:** Fails to integrate system components effectively • **0 pts - No Marks**	20 pts
Error Handling and Validation	• **20 pts - Full Marks:** Implements comprehensive error handling and data validation across all system components • **16 pts - Proficient:** Good error handling and validation with minor oversights • **12 pts - Competent:** Basic error handling and validation with some missing or ineffective implementations • **8 pts - Developing:** Inadequate error handling and validation with numerous issues • **4 pts - Beginning:** Fails to implement effective error handling and validation • **0 pts - No Marks**	20 pts
User Interface and Usability	• **20 pts - Full Marks:** Provides an intuitive, user friendly interface with excellent usability across all system components • **16 pts - Proficient:** Good user interface with minor usability issues or inconsistencies • **12 pts - Competent:** Adequate user interface with noticeable usability limitations • **8 pts - Developing:** Inadequate user interface with noticeable usability limitations • **4 pts - Beginning:** Poor user interface with significant usability issues. Fails to provide a usable and intuitive user interface • **0 pts - No Marks**	20 pts
Final Presentation	• **20 pts - Full Marks:** Delivers an outstanding final presentation • **16 pts - Proficient:** Good final presentation with minor areas lacking detail • **12 pts - Compentent:** Adequate presentation with some areas needing improvement • **8 pts - Developing:** Inadequate presentation with significant areas lacking clarity or detail • **4 pts - Beginning:** Fails to provide a coherent final presentation • **0 pts - No Marks**	20 pts
	Total	**80 pts**